PRATYUSH KUMAR SAHU

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PROFESSIONAL SUMMARY

Engineering student and full-stack developer experienced in designing, building, testing, and deploying production-style web applications (REST APIs, data modelling, authentication/security, and AI/RAG integration), with a track record of translating business requirements into technical outcomes and structured documentation across startup operations and applied research.

EDUCATION

Veer Surendra Sai University of Technology (VSSUT), Burla
Bachelor of Technology in Electrical Engineering | CGPA: 8.41/10 Expected Graduation: 2027

TECHNICAL PROJECTS

T&P Assistant: AI-Powered Placement Portal
React 19, Vite, Node.js/Express 5, MongoDB Atlas (incl. vector search), JWT, Gemini API

- Architected and built a full-stack web application end-to-end (React SPA frontend, Express REST API, MongoDB Atlas database) with JWT authentication (httpOnly refresh cookies, bcrypt hashing) and server-side, role-based access control for students and admin staff.
- Built a Retrieval-Augmented Generation (RAG) pipeline over MongoDB Atlas vector search and the Gemini API so the AI assistant answers only from admin-uploaded source documents, with cited sources rather than fabricated answers.
- Developed an AI-assisted resume analyzer (PDF parsing, rule-based ATS checks, AI feedback, JD-match scoring) and hardened the API for production with input validation, rate limiting, and secure headers (helmet).
- Delivered through an 8-phase, gated process: each phase implemented, tested, and documented (PRD, architecture, API reference, decision log) before the next began.

FlashPDF AI: AI-Generated Flashcard Study Platform
React 19/TypeScript, Node.js/Express, Prisma/SQLite, Gemini API, Docker

- Built a full-stack study app end-to-end (React/TypeScript frontend, Express backend, Prisma/SQLite database) with JWT-authenticated, per-user data isolation and an invite-gated signup flow.
- Implemented PDF text extraction (pdfjs-dist) with an OCR fallback (tesseract.js) for scanned/image-only PDFs, and AI flashcard generation via the Gemini API using structured JSON output.
- Validated the build with 24 automated backend tests (node:test/supertest) and shipped a Docker-based deployment, debugging real build issues (native module compilation, OpenSSL detection) found only by actually running the container.

PROFESSIONAL EXPERIENCE

Founder's Office Intern | Dahibara Express May 2026 – July 2026

- Partnered directly with the founder on strategic initiatives, coordinating cross-functional activities and market research to support targeted process improvements.

Outreach Intern | Craly.in April 2026 – June 2026

- Led lead generation and stakeholder outreach, applying structured market research and CRM tools to identify and prioritize business opportunities.

Research Intern: Electric Vehicle Integration | NITTTR Kolkata June 2025 – July 2025

- Researched EV integration and built MATLAB/Simulink models to analyze system performance; documented findings in structured technical reports for stakeholder review.

LEADERSHIP EXPERIENCE

Computer Science Mentor

- Mentored school students in Computer Science, building structured learning plans and simplifying technical concepts to fit individual needs.

Assistant Branch Secretary | Electrical Engineering Department, VSSUT

- Coordinated departmental events, volunteer teams, and project timelines; collaborated with faculty and student teams to plan and execute initiatives across department levels.

SKILLS

- **Programming & Scripting:** JavaScript, TypeScript, Python, C++, SQL, MATLAB/Simulink
- **Web & Backend Development:** React, Node.js/Express, REST API design, JWT/Auth, MongoDB, Prisma, Docker, Git
- **AI & Data:** RAG Pipelines, Vector Search, LLM API Integration (Gemini), AI-Assisted Development (Claude Code), Automated Testing
- **Project & Process:** Project Coordination, Stakeholder Management, Technical Documentation, Process Improvement, Agile Delivery

CERTIFICATIONS

- Generative AI Introduction (Google) – Link-to-Certificate
- Full Stack Web Development
- Google Network Architecture – Link-to-Certificate
- Project Management Plan using Trello – Link-to-Certificate
- Introduction to LLMs – Link-to-Certificate